TRIVIALITY OF THE CENTRE OF A LIE SUPERALGEBRA PASSES TO THE ENVELOPING ALGEBRA OF ITS POLYNOMIAL CURRENT ALGEBRA

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Allowed standard prerequisites (editorial).

- Reduction of the centre of the enveloping algebra to the invariants of the adjoint action, via the Poincare-Birkhoff-Witt theorem. Cited inside the counted proof at lines 1247-1252. Shows that triviality of the centre of U(a[u]) follows once the invariants of the adjoint action of a[u] on its supersymmetric algebra are trivial.
- A homogeneous basis of the Lie superalgebra and its structure constants. Used inside the counted proof at lines 1255-1261. Provides the coordinates

Date: doi:10.3842/SIGMA.2020.112; CC BY 4.0.

f_1, ..., f_K, their Z_2-degrees and the constants C_{pq}^r in which the whole computation is written.

Proposition 0.1. *Suppose the centre of the Lie superalgebra* $\mathfrak{a}$ *is trivial. Then the centre of the universal enveloping algebra* $\mathrm{U}(\mathfrak{a}[u])$ *is also trivial, that is equal to* $\mathbb{C}$.

Proof. Consider adjoint action of the Lie superalgebra $\mathfrak{a}[u]$ on its supersymmetric algebra. By Poincaré–Birkhoff–Witt theorem for $\mathrm{U}(\mathfrak{a}[u])$ it suffices to show that the space of invariants of this action is trivial, see [1, Theorem 5.15].

Let A be an element of the supersymmetric algebra of $\mathfrak{a}[u]$ invariant under the adjoint action. Let $K = \dim \mathfrak{a}$. Choose any $\mathbb{Z}_2$-homogeneous basis $f_1, \dots, f_K$ of $\mathfrak{a}$ and write the Lie brackets as

$$[f_p, f_q] = \sum_{r=1}^{K} C_{pq}^r f_r,$$

where $C_{pq}^r \in \mathbb{C}$ are the structure constants. Let d be the maximal degree such that A depends on at least one of the basis elements $f_1 u^d, \dots, f_K u^d$ of $\mathfrak{a}[u]$. Write

$$A = \sum_{m_1,\dots,m_K} A_{m_1\dots m_K} \big(f_1 u^d\big)^{m_1} \cdots \big(f_K u^d\big)^{m_K}, \tag{0.1}$$

where the coefficients $A_{m_1\dots m_K}$ are certain elements of the supersymmetric algebra which do not depend on $f_1 u^d, \dots, f_K u^d$. For each $p = 1, \dots, K$ let $\bar{p}$ be the $\mathbb{Z}_2$-degree of the basis element f_p of $\mathfrak{a}$. We allow $m_1, \dots, m_K$ in (0.1) to range over $0, 1, 2, \dots$ but assume that $A_{m_1\dots m_K} = 0$ if $m_p > 1$ for at least one index p with $\bar{p} = 1$.

For each $p = 1, \dots, K$ we have the equation $\mathrm{ad}(f_p u)(A) = 0$ in the supersymmetric algebra of $\mathfrak{a}[u]$. In particular, the component of the left-hand side of this equation that involves any of the basis elements $f_1 u^{d+1}, \dots, f_K u^{d+1}$ must be zero. Thus

$$\sum_{m_1,\dots,m_K} A_{m_1\dots m_K} \sum_{q,r=1}^{K} m_q C_{pq}^r \big(f_1 u^d\big)^{m_1} \cdots \big(f_q u^d\big)^{m_q - 1} \cdots \big(f_K u^d\big)^{m_K} f_r u^{d+1} (-1)^{h_q} = 0,$$

where we used the notation

$$h_q = \sum_{s=1}^{q-1} \bar{p}\bar{s} m_s + \sum_{s=q+1}^{K} \bar{r}\bar{s} m_s. \tag{0.2}$$

If follows that for any non-negative integers $m'_1, \dots, m'_K$ we have the equations

$$\sum_{q=1}^{M} A_{m'_1\dots m'_q+1\dots m'_K} (m'_q + 1) C_{pq}^r (-1)^{h'_q} = 0, \qquad \text{where} \quad p, r = 1, \dots, K. \tag{0.3}$$

Similarly to (0.2) here we used the notation

$$h'_q = \sum_{s=1}^{q-1} \bar{p}\bar{s} m'_s + \sum_{s=q+1}^{K} \bar{r}\bar{s} m'_s.$$

Let us now fix any non-negative integers $m'_1, \dots, m'_K$ and observe that the elements

$$f'_q = (m'_q + 1) f_q (-1)^{h'_q} \qquad \text{with} \quad q = 1, \dots, K$$

also form a basis of $\mathfrak{a}$. Since the centre of the Lie superalgebra $\mathfrak{a}$ is trivial, the system

$$\left[f_p, \sum_{q=1}^{K} w_q f'_q \right] = 0 \qquad \text{with} \quad p = 1, \dots, K$$

of linear equations on $w_1, \dots, w_K \in \mathbb{C}$ has only zero solution. This system can be written as

$$\sum_{q=1}^{K} w_q (m'_q + 1) C^r_{pq} (-1)^{h'_q} = 0, \qquad \text{where} \quad p, r = 1, \dots, K.$$

Hence by comparing the latter system with (0.3) we obtain that $A_{m'_1 \dots m'_q + 1 \dots m'_K} = 0$ for each index $q = 1, \dots, K$. It follows that $A \in \mathbb{C}$ and that $d = 0$ in particular. $\square$

References

[1] Milnor J.W., Moore J.C., On the structure of Hopf algebras, *Ann. of Math.* **81** (1965), 211–264.